

LECTURE ACTIVITY 9: SLEEPLESS NIGHTS – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Identify the population and the sample in a statistical study.
- 2) Identify whether a sample is representative of the population.
- 3) Apply the correct symbols/words and distinguish between statistic and parameter when referring to a quantitative variable.
- 4) State the null and the alternative hypotheses in words and in terms of the symbol μ , the population/long-run mean.
- 5) Predict the center, variation, and shape of the null distribution (distribution of sample means) from repeated random samples of the same size.
- 6) Use a simulation-based analysis to conduct a test involving the mean of a single quantitative variable.
- 7) State a conclusion about the alternative hypothesis (and null hypothesis) based on the p -value

Key Terms:

- | | | |
|---------------------|-----------------|-------------------------|
| • Quantitative data | • Sample mean | • Sample SD |
| • Population mean | • Population SD | • Sampling distribution |
-

Step 1: State the Research Question

How much do STAT 1110 students at your university sleep on a typical night? Let's make the question more specific and ask about last night. Did all STAT 1110 students have an average amount of sleep (in hours) less than the recommended eight hours?



Question 1

- (a) Identify the observational unit in this study.

Solution: One observational unit is one student.

- (b) Define the variable in this study. Is this variable quantitative or categorical? If it is categorical, what are its levels?

Solution: The variable is the **amount of sleep, in hours**, which is a **quantitative variable**.

Note: I recommend getting into the habit of including units in the definitions of quantitative variables.

- (c) Based on the research question, what is the population of interest?

Solution: The population of interest is the **set of all STAT 1110 students**.

- (d) Define the parameter of interest in words, and identify the correct symbol we use to denote this parameter. (This will require you to have completed the reading.)

Solution: The parameter of interest is the **population mean number of hours slept last night across all STAT 1110 students**. We use the symbol μ to denote this, since it is a *population* mean (as opposed to a *sample* mean, which would be denoted by $\bar{x}$).

- (e) State the null and alternative hypotheses in **symbols** and in **words**.

Solution:

- **In Words:** the null hypothesis is “the average amount of sleep last night across all STAT 1110 students is equal to 8” and the alternative is “the average amount of sleep last night across all STAT 1110 students is less than 8.”
- **In Symbols:** $H_0 : \mu = 8$ vs. $H_A : \mu < 8$.

Note: As discussed in lecture, your alternative and null hypotheses should never overlap - that is, there should never be a case where the null and alternative could both be simultaneously true. As such, it is **incorrect** to write the alternative as $H_A : \mu \leq 8$.

Step 2: Design a Study / Collect Data

To test the hypotheses we just formulated in Question 1(e), we need data!

Question 2

Ideally, how should we obtain a sample from the population of all STAT 1110 students?

Solution: Ideally, we would use an unbiased sampling method to obtain a sample of STAT 1110 students from the entire university. So, one idea would be to obtain a **simple random sample** of STAT 1110 students from the entire university. In this way, the sample would be representative of the population.

Whereas obtaining a simple random sample of STAT 1110 students is an ideal way to attempt to get a representative sample, there is no question that it would take a fair bit of work to make that happen. So much so, that we might not want to do the study anymore! This happens frequently in statistics, and so researchers often opt to use a **convenience sample** instead. One choice of a convenience sample of STAT 1110 students is to use our lecture section.

Question 3

- (a) Identify at least one way that our lecture section is likely a good representation of all STAT 1110 students at Cal Poly in terms of hours slept last night.

Solution: Answers may vary! You are all STAT 1110 students, so y'all are part of the population of interest. You also have similar age and academic responsibilities to other STAT 1110 students, and these variables might be related to the number of hours students sleep.

(b) Identify at least one way that our lecture section is likely a **not** good representation of all STAT 1110 students at Cal Poly in terms of hours slept last night.

Solution: Answers may vary! Students in this section may have different sleep habits than students in other sections because of the time of day this section is held. If some sections are held online (remotely), this might also play a role (e.g. with online students possibly sleeping longer, on average).

Recall from our previous discussion on sampling methods, convenience sampling may be biased whereas simple random sampling is an unbiased sampling method. We will revisit the implications of using our lecture section as a sample of STAT 1110 students later, when we evaluate the study. For now, let's gather the data.

Step 3: Explore the Data

We'll imagine that I sent a survey to my STAT 1110 students, and that 70 people responded back (I'll keep referring to this as "our lecture section"). The dotplot below summarizes the results of this survey.

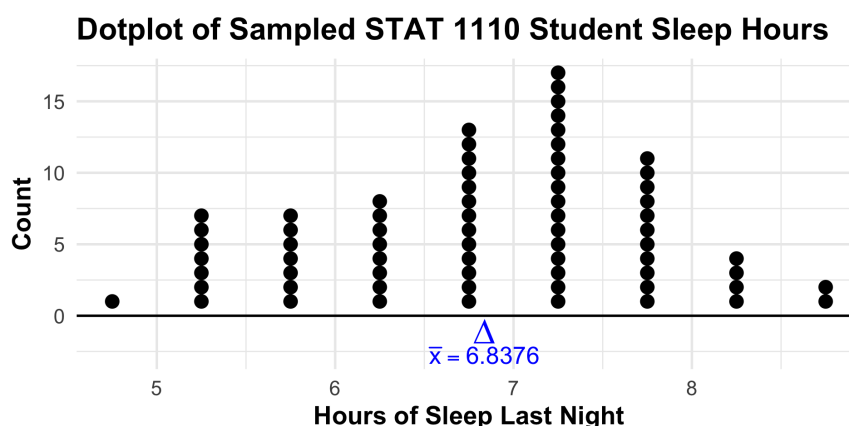


Figure 1: Results from a (fictitious) survey sent to 70 STAT 1110 students.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 6.84$	median = 7	$s = 0.93$	$n = 70$

Table 1: Summary statistics of the (fictitious) data.



Question 4

What does one dot on Figure (1) represent?

Solution: One dot represents one student and the amount they slept last night.

After we collect data, the next step is to explore it. In the case of a quantitative variable (like we have here), exploring the data is a bit more involved than in Chapter 1 (with categorical data). In particular, we'll need to refer back to our work from the Preliminaries chapter!

Recall that we can summarize the distribution of a quantitative variable (displayed either in a histogram or a dotplot) by describing **shape**, **center**, **variability**, and **outliers** (unusual observations). Here's one way to describe the distribution of sleep hours:

The distribution of hours of sleep for students in the sample is somewhat symmetric and bell-shaped centered around 6.84 which is a measure of the typical hours of sleep for this sample. The distribution varies from 4.5 hours to 9.5 hours of sleep, with a standard deviation of 0.93 which means that the average distance from each of the hours of sleep to the overall mean (6.84) is about 0.93.



Question 5

Does the data provide any *preliminary* evidence that Cal Poly students tend to sleep less than 8 hours a night?

Solution: Yes, we do have preliminary evidence. We can see some variation in student's answers but the sample mean of 6.84 hours of sleep in the previous night is well below the hypothesized mean of 8 hours, which might suggest students may sleep less than 8 hours on average. We do not know the answer for sure yet because we do not know if these results could have happened by chance. But it does seem like it.

Step 4: Draw Inferences Beyond the Data

Now that we've explored the data and better understand the distribution of last night's sleep hours in this class, we will learn how to use both **simulation-based** (today) and **theory-based** (next time) approaches to evaluate the strength of the evidence against the null hypothesis.

Concept Check-In

Recall that in our explorations of categorical data (Lecture Activities 5 through 8), we found it crucial to distinguish the different types of "proportions" we saw: in particular, distinguishing *sample* proportions (which we denote by $\hat{p}$) from population (or long-run) proportions (which we denote by π).

Even with quantitative data, it's important to distinguish sample-based quantities from population-based (or process-based, if we're in the realm of random processes) quantities. Just like with quantitative data, we have separate notation we use to highlight these distinctions:

Quantity	Statistic or Parameter?	Symbol
Sample Mean	Statistic	$\bar{x}$
Population Mean	Parameter	μ
Sample Standard Deviation	Statistic	s
Population Standard Deviation	Parameter	σ

Table 2: Notations used for statistics and parameters, with quantitative data.

Question 6

Fill in the middle column of Table (2) above, to indicate whether each specified quantity is a statistic or a parameter.

Solution: See table above.

Caution

We will expect you to use the correct notation going forward! Remember - this is not just “something we do for fun-” using the correct notation indicates to your reader that you understand the crucial difference between statistics and parameters.

Question 7

Suppose we had taken many, many samples of the same size from the population. If the null hypothesis is true, how do you predict the distribution of sample means will behave? Be sure to discuss any information you do not know.

Solution: Recall that the null hypothesis is (in words): “the average amount of sleep last night among all STAT 1110 students is 8 hours.” As such, we’d expect samples of STAT 1110 students to have *sample* average sleep hours of around 8 as well - that is to say, **we’d expect the distribution of sample means to be centered at around 8**. But, there will also be some variation - some sample means will fall above 8, and others will fall below 8. **At this stage, we don’t really know what the variability would look like, though.**

As for shape, recall our sampling from the Gettysburg Address (Lecture Activity 3) - based on our findings from that activity, we might expect the distribution of sample means to be **unimodal and bell-shaped**.

3S Strategy

Just like we did before (with categorical data), we can implement the **3S** strategy to conduct a simulation-based test of the hypotheses postulated in Question 1(e).

Statistic

Question 8

Remind yourself of the sample mean sleep hours last night from the dataset above:

- $\bar{x} =$ 6.84

Simulate

Simulating with quantitative data is a little different than simulating with categorical data, mainly because we don’t have a nice analogy like “tossing a coin” or “spinning a spinner” to represent a simulated observational unit any more. Instead, we typically:

- 1) Manufacture a large, hypothetical population of many thousands of units (in this case, college students) under the null
- 2) Take simple random samples from this hypothetical population

We need to know what *mean* and *standard deviation* to use when manufacturing our “large, hypothetical population.” The mean part is easy (you should know why now! Hint: we’re doing everything *under the null*), but it’s not so clear what to put for the population standard deviation. For now, let’s assume $\sigma = 1.5$, just so we can proceed with our inference.

The applet that will enable us to do this is the **One Mean Applet**. When you navigate to the applet, you should see “Sleep 1” has been selected by default and a histogram displays its distribution. This is the **population distribution** - namely, the distribution of values within the population.

Caution

Remember:

- a) This is a **simulated (hypothetical)** population distribution (**that we simulated assuming the null hypothesis is true**). It does NOT actually represent the sleep habits of all STAT 1110 students last night.
- b) In general, we will never get to see the true population distribution. Because, seeing the population distribution requires us to have all values from the population, meaning there’s no need to conduct inference - the value of μ is completely known! (Which, again, is not typically the case in real-world problems.)



Question 9

Navigate to the One Mean applet. Then, fill in the blanks below

- The mean of the simulated/hypothetical population distribution is 8.001.
- The SD of the simulated/hypothetical population distribution is 1.510.
- The size of the simulated/hypothetical population distribution is 18,000.

It’s time to start simulating the null distribution! Remember: the null distribution is just the distribution of statistic values (of a particular statistic), under the null. For example: in this study our statistic of interest is the *sample mean* $\bar{x}$ meaning we’ll need to simulate the null distribution of $\bar{x}$.



Question 10

Navigate to the One Mean applet. Then:

- 1) Check the “Show Sampling Options” box
- 2) Keep “Number of Samples” set to 1 for now
- 3) Set the “Sample Size” to match the class data from above
- 4) Press “Draw Samples”
- 5) Right below the “Draw Samples” button, see the ID numbers and sleep time values that were randomly selected from this hypothetical population.

Record the sample mean for this sample from the “Most Recent Sample” dotplot. Notice that this sample mean is added to the “Statistic” dotplot. **Don’t close out of the plot!**

- My observed sample mean is 8.107 (Answers may vary)

Question 11

Navigate back to the One Mean applet. Then:

- 1) Change the “Number of Samples” to 9,999 (to obtain a total of 10,000 simulated samples, including our initial sample from the previous question).
- 2) Press “Draw Samples.”

Include a screenshot of **your entire screen** below, showing all inputs and options you specified along with any relevant outputs. **Don’t close out of the plot yet!**

Solution: Answers may vary; here’s my plot:

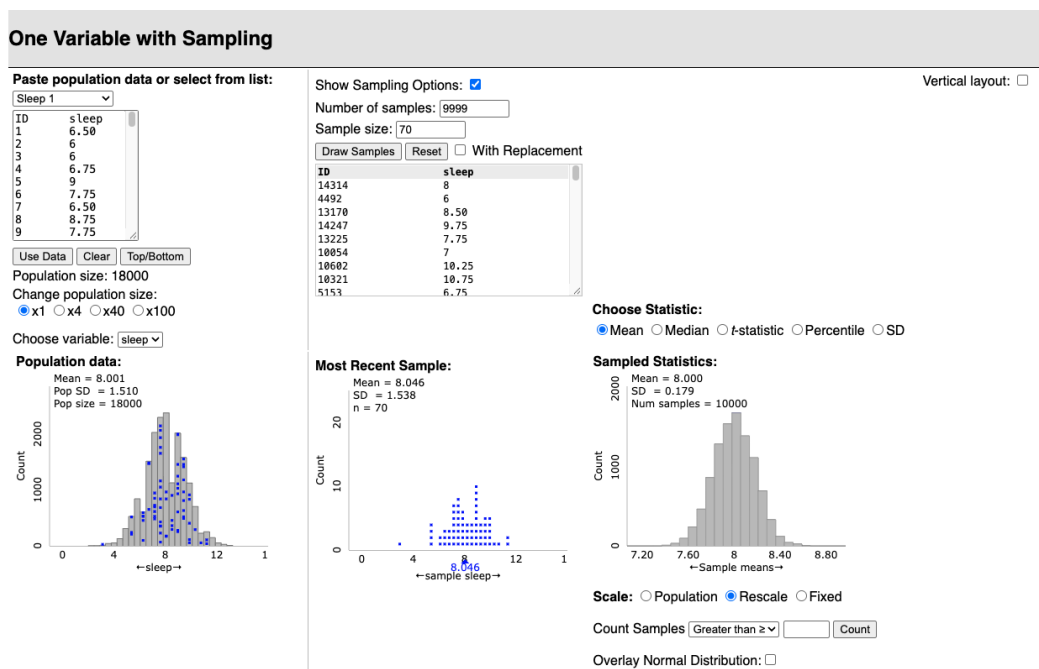


Figure 2: Results from the One Mean Applet.

Question 12

Refer back to the histogram (of sample means) you just created using the One Mean Applet.

- (a) What is the shape of the distribution of sample means?

Solution: The distribution of sample means is approximately bell-shaped and symmetric.

- (b) Where is the distribution of sample means centered? Why does this make sense?

Solution: The distribution of sample means is centered around 8 hours because the random samples are being randomly chosen from a population with a mean of 8 hours (this was the null hypothesis assumption: $\mu = 8$).

- (c) What is the standard deviation of the distribution of sample means (SD_{NULL})?

Solution: Answers may vary slightly, but they should all be around 0.179

Strength of Evidence

Question 13

Find and report the approximate p -value. **To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) p -value.**

- My approximate p -value is 0.000

Here is my screenshot:

Solution: Answers may vary; here's my plot:

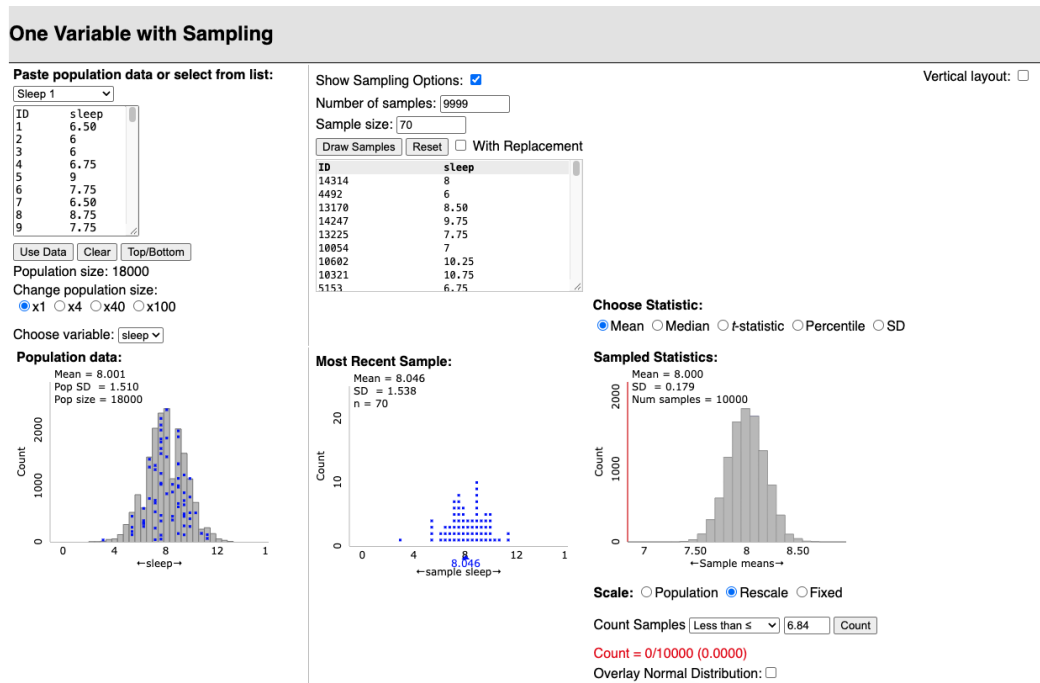


Figure 3: Results from the One Mean Applet.

Question 14

Evaluate this p -value. How much evidence does the sample data provide against the null hypothesis? Choose one.

- ☐ Not much evidence against the null hypothesis
- ☐ Moderate evidence against the null hypothesis
- ☐ Strong evidence against the null hypothesis
- ☒ **Very strong evidence against the null hypothesis**



Question 15

Interpret this p -value.

Solution: The p -value of 0.000 is the probability of observing a sample mean of 6.84 hours of sleep in the previous night or less, assuming the true population mean sleep time is 8 hours (null hypothesis is true).

Recall that an alternate measure of strength of evidence is the **standardized statistic**, which we previously defined as

$$z = \frac{(\text{observed statistic value}) - (\text{hypothesized mean of null distribution})}{(\text{hypothesized SD of null distribution})}$$

This definition works equally well for a sample mean as our sample statistic, as it did when our statistic was a sample proportion!



Question 16

Calculate the value of the standardized statistic (recall that our original sample of STAT 1110 students had $\bar{x} = 6.84$), using the hypothesized mean and SD of the null distribution you obtained from the One Mean Applet.

Solution: My hypothesized mean of the null distribution was 8.000 and my hypothesized standard deviation of the null distribution was 0.179, meaning the standardized statistic is

$$z = \frac{6.84 - 8.000}{0.179} \approx \mathbf{-6.481}$$

Even though individual answers may vary slightly (do you understand why?), I suspect most of you got a standardized statistic that was fairly large in magnitude (absolute value). As such, our observed statistic is very far in the tails of the (simulated) null distribution, meaning it provides very strong evidence against the null hypothesis.

Step 5: Formulate Conclusions

As we progress further into the course, it will become more and more important for us to spend some time thinking about our conclusions. Let's start getting some practice with that!

Our research question was: is the average hours of sleep for STAT 1110 students less than the recommended eight hours?

 Question 17

Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p-value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p-value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the parameter in your conclusion.

Solution: A very small p -value of 0.000 was obtained based on the statistic of 6.84 hours (sample mean hours of sleep in the previous night). This constitutes very strong evidence against the null hypothesis. Therefore, we have very strong evidence to reject the null hypothesis and conclude that the population mean sleep time is less than 8 hours, for STAT 1110 students like the ones in the study.

 Question 18

To which population can you apply the results of this study? Try to be as specific as you can in your answer to this question. As a hint, just saying “STAT 1110 students” is not quite sufficient.

Solution: The results could be applied to STAT 1110 students who are similar to those in the sample. Possibly STAT 1110 students with similar workloads, similar ages, and taking classes at similar times.

Step 6: Look Back and Ahead

 Question 19

Critique this study. (This is intentionally a bit open-ended!)

Solution: One limitation is that students self-reported their sleep hours, which may introduce error. Another limitation is that the sample was a convenience sample from a single STAT 1110 class rather than a random sample of all STAT 1110 students.